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DISPLAY PANEL AND DISPLAY DEVICE

Abstract

A display panel and a display device are provided. The display panel includes: a display region and a non-display region, the non-display region surrounds the display region, the display region includes a first display region and a second display region, the first display region surrounds the periphery of the second display region, a plurality of first pixels is arranged in the first display region, a plurality of second pixels is arranged in the second display region, the display region includes a first driving circuit configured to drive the first pixels and a second driving circuit configured to drive the second pixels; and one first driving circuit is configured to simultaneously drive at least two first pixels.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Chinese Patent Application No. 202211204118.7 filed in China on Sep. 29, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of display technologies, in particular to a display panel and a display device.

BACKGROUND

[0003] Nowadays, there are more and more electronic products, such as mobile phones, televisions, tablets and wearable products. With the development of electronic products, a display technology has been widely used. The functions and usage scenarios of display products are becoming more numerous, leading to higher demands on the display area of display devices. In a display device, the wider a border of the device is, the larger a non-display region is, and the smaller an area of an actual display region is. For some miniaturized display panels, such as various types of special-shaped wearable products, for example, a circular-shaped display device with rounded corners, the border of the non-display region is relatively wide due to the presence of rounded corners, so that an area of the display region is small and a screen-to-body ratio is low, and it is difficult to meet the requirements on the narrow border of the display product, thereby adversely affecting the user's experience.

SUMMARY

[0004] An object of embodiments of the present disclosure is to provide a display panel and a display device.

[0005] In a first aspect, the embodiments of the present disclosure provide a display panel, including a display region and a non-display region. The non-display region surrounds the display region, the display region includes a first display region and a second display region, the first display region surrounds the periphery of the second display region, a plurality of first pixels is arranged in the first display region, a plurality of second pixels is arranged in the second display region, the display region includes a first driving circuit configured to drive the first pixels and a second driving circuit configured to drive the second pixels, and one first driving circuit is configured to simultaneously drive at least two first pixels.

[0006] Optionally, one first driving circuit is configured to simultaneously drive two first pixels in two adjacent columns respectively.

[0007] Optionally, two first pixels respectively in two columns driven simultaneously by a same first driving circuit are in a same row.

[0008] Optionally, at least part of the first driving circuit is arranged in the second display region.

[0009] Optionally, the display panel further includes: connection lines, each of the connection lines is connected to electrodes of at least two first pixels, and each first driving circuit is electrically connected to a corresponding one of the connection lines.

[0010] Optionally, each of the connection lines is connected to two adjacent first pixels.

[0011] Optionally, each of the first pixels includes a first sub-pixel, and one first driving circuit is configured to simultaneously drive first sub-pixels of at least two first pixels.

[0012] Optionally, each of the first pixels includes a second sub-pixel, and one first driving circuit is configured to simultaneously drive second sub-pixels of at least two first pixels, where the first sub-pixel and the second sub-pixel emit light in different colors.

[0013] Optionally, the first pixel includes: a first electrode, a light-emitting unit and a second electrode laminated one on another, and one first driving circuit is configured to connect electrodes in at least two first pixels.

[0014] Optionally, the display panel further includes: a detection circuit configured to detect brightness of a pixel, at least part of the detection circuit being located in the first display region.

[0015] Optionally, the display panel further includes: a branching unit, and the first driving circuit is electrically connected to the first pixels via the branching unit.

[0016] Optionally, at least part of the branching unit is located in the first display region.

[0017] Optionally, the first display region includes a first region, a second region, a third region and a fourth region, the first region, the second region, the third region and the fourth region are distributed along the periphery of the second display region, and the first pixels are arranged in at least one of the first region, the second region, the third region and the fourth region.

[0018] Optionally, a distribution density of the first pixels in the first display region is less than a distribution density of the second pixels in the second display region.

[0019] In a second aspect, the embodiments of the present disclosure provide a display device including the display panel described in the above embodiments; and the display device further includes: a frame body covering the non-display region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1a is a schematic view showing a distribution of a display region and a non-display region according to the embodiments of the present disclosure;

[0021] FIG. 1b is another schematic view showing the distribution of the display region and the non-display region in an embodiment of the present disclosure;

[0022] FIG. 1c is yet another schematic view showing the distribution of the display region and the non-display region according to the embodiments of the present disclosure;

[0023] FIG. 1d is still yet another schematic view showing the distribution of the display region and the non-display region according to the embodiments of the present disclosure;

[0024] FIG. 1e is still yet another schematic view showing the distribution of the display region and the non-display region according to the embodiments of the present disclosure;

[0025] FIG. 1f is still yet another schematic view showing the distribution of the display region and the non-display region according to the embodiments of the present disclosure;

[0026] FIG. 1g is still yet another schematic view showing the distribution of the display region and the non-display region according to the embodiments of the present disclosure;

[0027] FIG. 2 is a schematic view showing a connection between a connection line and a pixel region according to the embodiments of the present disclosure;

[0028] FIG. 3 is a schematic view showing a connection between a connection line and electrodes according to the embodiments of the present disclosure;

[0029] FIG. 4 is another schematic view showing the connection between the connection line and the pixel region according to the embodiments of the present disclosure;

[0030] FIG. 5 is another schematic view showing the connection between the connection line and the electrodes according to the embodiments of the present disclosure;

[0031] FIG. 6 is a schematic view showing a case where a connection line is connected to

electrodes and a first driving circuit according to the embodiments of the present disclosure;
[0032] FIG. 7 is another schematic view showing the connection between the connection line and the pixel region according to the embodiments of the present disclosure;
[0033] FIG. 8 is a schematic view showing a connection between a connection line and a pixel according to the embodiments of the present disclosure;
[0034] FIG. 9 is another schematic view showing the connection between the connection line and the pixel according to the embodiments of the present disclosure;
[0035] FIG. 10 is yet another schematic view showing the connection between the connection line and the pixel according to the embodiments of the present disclosure; and
[0036] FIG. 11 is still yet another schematic view showing the connection between the connection line to the pixel according to the embodiments of the present disclosure.

REFERENCE SIGN LIST

[0037] **10** display region [0038] **11** first display region [0039] **12** second display region [0040] **111** first region [0041] **112** second region [0042] **113** third region [0043] **114** fourth region [0044] **20** non-display region [0045] **21** border region [0046] **30** first pixel [0047] **31** first sub-pixel [0048] **32** second sub-pixel [0049] **33** third sub-pixel [0050] **40** second pixel [0051] **51** first driving circuit [0052] **52** second driving circuit [0053] **60** connection line [0054] **61** branching unit [0055] **62** pixel region

DETAILED DESCRIPTION

[0056] The technical solutions in the embodiments of the present disclosure will be described hereinafter clearly and completely with reference to the drawings of the embodiments of the present disclosure. Apparently, the following embodiments merely relate to a part of, rather than all of, the embodiments of the present disclosure, and based on these embodiments, a person of ordinary skill in the art may, without any creative effort, obtain other embodiments, which also fall within the scope of the present disclosure.

[0057] Terms such as “first” and “second” in the description and the claims of the present disclosure are used to distinguish similar objects and are not necessarily used to describe a specific order or sequence. It should be understood that the terms used in this way may be interchanged under appropriate circumstances, so that the embodiments of the present disclosure described herein may be implemented in a sequence other than those illustrated or described herein. In addition, in the description and the claims, “and/or” means at least one of the connected objects, and the character “/” generally means that the associated objects are in an “or” relationship.

[0058] Hereinafter, a display panel according to the embodiments of the present disclosure will be described in detail with reference to FIGS. 1a to 11 through specific embodiments and application scenarios thereof.

[0059] As shown in FIGS. 1a to 11, the display panel of the embodiments of the present disclosure includes: a display region **10** and a non-display region **20** surrounding the display region **10**, and the display panel may have a circular shape or a rectangular shape. The display panel may be a non-flexible panel, and may be applied to a mobile phone, a tablet, a computer, and a wearable device such as a watch. The display region **10** includes a first display region **11** and a second display region **12**, the first display region **11** surrounds the periphery of the second display region **12**, a plurality of first pixels **30** is arranged in the first display region **11**, a plurality of second pixels **40** is arranged in the second display region **12**, light-emission colors of the first pixels **30** and the second pixels **40** may be the same or different, and a distribution density or arrangement manner of the first pixels **30** may be the same with, or different from, a distribution density or arrangement manner of the second pixels **40**. The display region **10** includes a first driving circuit **51** configured to drive the first pixels **30** and a second driving circuit **52** configured to drive the second pixels **40**, one first driving circuit **51** is configured to drive at least two first pixels **30** simultaneously. The first display region **11** may have a pixel region **62**, each first pixel **30** may be arranged in the corresponding pixel region **62**, and the first driving circuit **51** may be connected to the first pixel **30**

through a via hole of the pixel region **62**. One second driving circuit **52** may drive one or more second pixels **40**, for example, each second driving circuit **52** may drive one second pixel **40**. At least two first pixels **30** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** may be driven simultaneously by one first driving circuit **51**. When one first driving circuit **51** simultaneously drives two first pixels **30**, it is able to reduce the quantity of first driving circuits **51**, thereby to reduce the space occupied by the first driving circuits **51**. The saved space may be used to accommodate other lines or devices, which is beneficial to increasing an area of the display region, and narrowing a border of the display panel.

[0060] As shown in FIGS. **1a** to **1g**, the display region **10** includes a first display region **11** and a second display region **12**, the first display region **11** surrounds the periphery of the second display region **12**, the first display region **11** may extend along the periphery of the second display region **12**, and the display region **10** may be a circular-shaped display region, an elliptical-shaped display region, a polygonal-shaped display region or a display region with rounded corners. The second display region **12** may be a circular-shaped display region, an elliptical-shaped display region, a polygonal-shaped display region or a display region with rounded corners. The first display region **11** may be ring-shaped, frame-shaped, arc-shaped or long strip-shaped. Specific shapes of the first display region **11** and the second display region **12** may be determined according to actual needs, and shapes and sizes of the first display region **11** and the second display region **12** may be matched according to practical applications.

[0061] For example, in the display panel, as shown in FIGS. **1a** to **1b**, the display region **10** of the display panel may be circular-shaped, the second display region **12** may be circular-shaped, the first display region **11** may be ring-shaped, and the first display region **11** is located at a peripheral region of the display region. For example, in the display panel, the display region **10** of the display panel may have an elliptical shape, the second display region **12** may have an elliptical shape, the first display region **11** may have a ring shape, and the first display region **11** is located at a peripheral region of the display region. For example, in the display panel, the display region **10** of the display panel may have an elliptical shape, the second display region **12** may have an elliptical shape, the first display region **11** may have a ring shape, and the first display region **11** is located at a peripheral region of the display region. For example, in the display panel, the display region **10** of the display panel may have a rectangular shape, the second display region **12** may have a rectangular shape, the first display region **11** may have a frame shape, and the first display region **11** is located at a peripheral region of the display region.

[0062] In some embodiments, as shown in FIGS. **1c** and **1f**, the display region **10** of the display panel may be rectangular-shaped, the second display region **12** may be rectangular-shaped, the first display region **11** may be frame-shaped, and the first display region **11** is located at the peripheral region of the display region.

[0063] The first display region **11** may include a first region **111**, a second region **112**, a third region **113** and a fourth region **114**, as shown in FIGS. **1b** and **1c**, the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may be successively distributed along the periphery of the second display region **12**. The first region **111**, the second region **112**, the third region **113** and the fourth region **114** may be sequentially in contact with and connected to encircle and form the first display region **11**, adjacent ones of the first region **111** and the second region **112**, the third and fourth regions **113**, **114** may be spaced apart from each other without contact, the first and third regions **111**, **113** may be arranged opposite to each other, and the second and fourth regions **112**, **114** may be arranged opposite to each other.

[0064] For example, as shown in FIG. **1b**, in the case where the first display region **11** has a ring shape, the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may each have an arc shape, the first region **111**, the second region **112**, the third region **113**, the fourth region may be concentrically distributed as the second display region **12**, and the first display region **11** may be formed by the first region **111**, the second region **112**, the third region **113**

and the fourth region that are sequentially in contact with and connected to.

[0065] As shown in FIG. 1c, in the case where the first display region **11** is of a frame shape, the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may each be of a strip-like shape, or the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may each be of a rectangular shape, the first region **111** and the third region **113** may be arranged in parallel and spaced apart from each other, the second region **112** and the fourth region **114** may be distributed in parallel and spaced apart from each other, the first region **111** and the second region **112** may be distributed vertically, and the third region **113** and the fourth region **114** may be distributed vertically.

[0066] In the display panel, as shown in FIG. 1d, the display region **10** of the display panel may be a display region having rounded corners, the display region **10** of the display panel may be a display region having one or more rounded corners, for example, the display region **10** may be substantially quadrangular-shaped, the display region **10** may have four rounded corners, the first display region **11** may be located at a peripheral region of the display region, the rounded corners may be located at corners of the display region, and the rounded corners may be located in the first display region **11**. As shown in FIG. 1e, adjacent ones of the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may be spaced apart from each other, the first region **111** and the third region **113** may each be located at a corner of the display region, the second region and the fourth region **114** may each be located at another corner of the display region, the first region **111** and the third region **113** may be arranged opposite to each other, the second region **112** and the fourth region **114** may be arranged opposite to each other, and the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may each be in an arc shape, so as to enable the corners of the display panel to each be of a rounded corner shape. For example, the first region **111** may be located in a lower left corner region of the display region, the second region **112** may be located in an upper left corner region of the display region, the third region **113** may be located in an upper right corner region of the display region, and the fourth region **114** may be located in a lower right corner region of the display region.

[0067] At least one of the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may have the first pixels **30**, for example, as shown in FIG. 1c, the first region **111** may be located at a left peripheral region of the display region, each first pixel **30** located in the first region **111** may be arranged in a corresponding pixel region **62**, and the first driving circuit **51** may be connected to the first pixel **30** in the first region **111** through a via hole of the pixel region **62**. At least two first pixels **30** located in the first region **111** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** located in the first region may be driven simultaneously by one first driving circuit **51**. When one first driving circuit **51** simultaneously drives two first pixels **30** located in the first region **111**, it is able to reduce the quantity of first driving circuits **51** required for the first pixels **30** located in the first region **111**, thereby to reduce the space occupied by the first driving circuits **51**. The saved space may be used to accommodate other lines or devices, which is beneficial to narrowing the border of the display panel close to the first region **111**.

[0068] As shown in FIG. 1c, the second region **112** may be located at an upper peripheral region of the display region, each first pixel **30** located at the second region **112** may be arranged in a corresponding pixel region **62**, and the first driving circuit **51** may be connected to the first pixel **30** in the second region **112** through a via hole of the pixel region **62**. At least two first pixels **30** located in the second region **112** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** located in the second region **112** may be driven simultaneously by one first driving circuit **51**. When one first driving circuit **51** simultaneously drives two first pixels **30** located in the second region **112**, it is able to reduce the quantity of first driving circuits **51** required for the first pixels **30** located in the second region **112**, thereby to reduce the space occupied by the first driving circuits **51**. The saved space may be used to accommodate other lines or devices,

which is beneficial to narrowing the border of the display panel close to the second region **112**.

[0069] As shown in FIG. **1c**, the third region **113** may be located at a right peripheral region of the display region, each first pixel **30** located at the third region **113** may be arranged in a corresponding pixel region **62**, and the first driving circuit **51** may be connected to the first pixel **30** in the third region **113** through a via hole of the pixel region **62**. At least two first pixels **30** located in the third region **113** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** located in the third region **113** may be driven simultaneously by one first driving circuit **51**. When one first driving circuit **51** simultaneously drives two first pixels **30** located in the third region **113**, it is able to reduce the quantity of first driving circuits **51** required for the first pixels **30** located in the third region **113**, thereby to reduce the space occupied by the first driving circuit **51**. The saved space may be used to accommodate other lines or devices, which is beneficial to narrowing the border of the display panel close to the third region **113**.

[0070] As shown in FIG. **1c**, the fourth region **114** may be located at a lower peripheral region of the display region, each first pixel **30** located at the fourth region **114** may be arranged in a corresponding pixel region **62**, and the first driving circuit **51** may be connected to the first pixel **30** in the fourth region **114** through a via hole of the pixel region **62**. At least two first pixels **30** located in the fourth region **114** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** located in the fourth region **114** may be driven simultaneously by one first driving circuit **51**. When one first driving circuit **51** simultaneously drives two first pixels **30** located in the fourth region **114**, it is able to reduce the quantity of first driving circuits **51** required for the first pixels **30** located in the fourth region **114**, thereby to reduce the space occupied by the first driving circuit **51**. The saved space may be used to accommodate other lines or devices, which is beneficial to narrowing the border of the display panel close to the fourth region **114**.

[0071] The display panel may have a circular shape, an elliptical shape, a polygonal shape, a shape structure with rounded corners, etc. and the display region of the display panel may have a circular shape, an elliptical shape, a polygonal shape, a shape structure with rounded corners, etc. and the display region may match the display panel in shape. In the application process, according to practical needs, at least two first pixels **30** among the first pixels **30** of at least one of the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may be driven simultaneously by one first driving circuit **51**, for example, two first pixels **30** may be driven simultaneously by one first driving circuit **51**. Thus, the quantity of first driving circuits **51** required for the first pixels **30** in at least one of the first region **111**, the second region **112**, the third region **113** and the fourth region **114** may be reduced, the space occupied by the first driving circuits **51** is also reduced, and the saved space may be used to accommodate other lines or devices, which is advantageous for narrowing the border of the corresponding region of the display panel.

[0072] As shown in FIG. **1f**, the display region **10** may include a first display region **11** and a second display region **12**. The first display region **11** surrounds the periphery of the second display region **12**, a plurality of first pixels **30** is arranged in the first display region **11**, a plurality of second pixels **40** is arranged in the second display region **12**, and a distribution density of the first pixels **30** may be the same with, or different from, a distribution density of the second pixels **40**. The first pixels **30** may be evenly distributed and spaced apart from each other, and the second pixels **40** may be evenly distributed and spaced apart from each other. The distribution density of the first pixels **30** in the first display region **11** may be less than the distribution density of the second pixels **40** in the second display region **12**. Since the distribution density of the first pixels **30** in the first display region **11** is less than the distribution density of the second pixels **40** in the second display region **12**, the quantity of first driving circuits required for driving the first pixels **30** in the first display region **11** is reduced, the space occupied by the first driving circuits **51** is reduced, and the saved space may be used to accommodate other lines or devices, which is beneficial to increasing the area of the display region and narrowing the border of the display panel.

[0073] As shown in FIG. 1g, the distribution density of the first pixels 30 in at least one of the first region 111, the second region 112, the third region 113 and the fourth region 114 is less than the distribution density of the second pixels 40 in the second display region 12. For example, the distribution density of the first pixels 30 in the first region 111 is less than the distribution density of the second pixels 40 in the second display region 12. Since the distribution density of the first pixels 30 in the first region 111 is less than the distribution density of the second pixels 40 in the second display region 12, the quantity of the first driving circuits required for driving the first pixels 30 in the first region 111 is reduced, the space occupied by the first driving circuits 51 is reduced, and the saved space may be used to accommodate other lines or devices, which is advantageous for increasing the area of the display region and for narrowing the border of the display panel narrower.

[0074] As shown in FIG. 1g, in each of the first region 111, the second region 112, the third region 113 and the fourth region 114, the distribution density of the first pixels 30 is less than the distribution density of the second pixels 40 in the second display region 12, the brightness of the first pixels 30 is greater than the brightness of the second pixels 40 in the second display region 12, so as to reduce a brightness difference caused by the distribution density of the first pixels 30 being less than the distribution density of the second pixels 40 in the second display region 12. Therefore, the brightness difference between the first pixels 30 and the second pixels 40 with different distribution densities is not easily observed, thereby improving the display effect and improving the viewing experience of a user.

[0075] The second display region 12 may include a peripheral region and a central region, the peripheral region may surround the periphery of the central region, the peripheral region is located between the first display region 11 and the central region, and distribution densities of the second pixels 40 in the peripheral region and the central region may be the same or different. For example, the distribution densities of the second pixels 40 in the peripheral region and the central region may be different, the distribution density of the second pixels 40 in the peripheral region may be less than the distribution density of the second pixels 40 in the central region, and the distribution density of the first pixels 30 in at least one of the first region 111, the second region 112, the third region 113 and the fourth region 114 is less than the distribution density of the second pixels 40 in the peripheral region. For example, the distribution density of the first pixels 30 in the first region 111 is less than the distribution density of the second pixels 40 in the peripheral region. The peripheral region may serve as a transition region, and the brightness difference caused by the distribution density of the first pixels 30 in the first display region 11 being less than the distribution density of the second pixels 40 in the second display region 12 may not easily observed through the second pixels 40 in the peripheral region, thereby improving the display effect and improving the viewing experience of the user.

[0076] The central region may be a circular-shaped display region, an elliptical-shaped display region, a polygonal-shaped display region, and a display region with rounded corners. The peripheral region may be ring-shaped, frame-shaped, arc-shaped or strip-shaped. Specific shapes of the peripheral region and the central region may be determined according to actual needs, and shapes and sizes of the peripheral region and the central region may be matched according to practical applications.

[0077] One first driving circuit 51 may drive at least two first pixels 30 simultaneously, one second driving circuit 52 may drive one or more second pixels 40, e.g., one first driving circuit 51 may drive two first pixels 30 simultaneously, each second driving circuit 52 may drive one second pixel 40. At least two first pixels 30 may be driven simultaneously by one first driving circuit 51, at least part of the first driving circuit 51 may be located in the second display region 12, and at least part of the first driving circuit 51 may be located in the peripheral region or the central region of the second display region 12, so as to reduce the space occupied by the first driving circuit 51 in the first display region 11. Hence, the saved space may be used to accommodate other lines or devices,

which is beneficial to increasing the area of the display region and narrowing the border of the display panel.

[0078] In the display panel of the embodiments of the present disclosure, the display region **10** includes the first driving circuit **51** configured to drive the first pixel **30** and the second driving circuit **52** configured to drive the second pixel **40**, one first driving circuit **51** is used to drive at least two first pixels **30** simultaneously. In the first display region **11**, one first driving circuit **51** is used to simultaneously drive at least two first pixels **30**, so it is able to reduce the quantity of the first driving circuits, thereby to reduce the space occupied by the first driving circuits. Hence, a part of the space may be saved in the peripheral region, and some other traces may be arranged in the saved space, so that the border of the display panel may be narrower, thereby to increase the area of the display region, improve the screen-to-body ratio, meet the requirement on the narrow border of the display product, and improve the user experience.

[0079] In some embodiments, when one first driving circuit **51** is used to drive two first pixels **30** in two columns simultaneously, it may reduce the difference in gray scale, thereby to avoid the occurrence of serrated display in the periphery and provide more uniform display. One first driving circuit **51** is configured to simultaneously drive two first pixels **30** in two columns respectively, and the two first pixels **30** in the two columns respectively may be in two adjacent columns respectively, such as a first column and a second column. The two first pixels **30** in the two columns respectively may be in two non-adjacent columns respectively, for example, the two first pixels **30** in the two columns respectively may be in the first column and the third column, or the two first pixels **30** in the two columns respectively may be in the first column and the fourth column. One first driving circuit **51** drives the two first pixels **30** in the two columns simultaneously, so as to facilitate the connection, which is beneficial to reducing the length of each of the lines through which the first driving circuit **51** is connected to the two first pixels **30**. The first pixels **30** may be arranged in an array form, one first driving circuit **51** may simultaneously drive two first pixels **30** in two columns respectively. The two first pixels **30** in two columns respectively driven simultaneously by the same first driving circuit **51** may be located in a same row, so as to provide a relatively small length difference between the lines through which the first driving circuit **51** is connected to two first pixels **30**, thereby to further reduce the length of each of the lines through which the first driving circuit **51** is connected to two first pixels **30**. One first driving circuit **51** is configured to simultaneously drive two first pixels **30** in two adjacent columns respectively, and the first pixels **30** may be arranged in an array form. When one first driving circuit **51** simultaneously drives two first pixels **30** in two columns respectively, two first pixels **30** in two columns respectively which are driven at the same time by the same first driving circuit **51** may be located in the same row, it is able to provide a relatively small length difference between the lines through which the first driving circuit **51** is connected to two first pixels **30**, thereby to further reduce the length of each of the lines through which the first driving circuit **51** is connected to two first pixels **30**, and facilitating the connection between the first driving circuit **51** and the first pixels **30**. In addition, it is also able to reduce the space occupied by the lines through which the first driving circuit **51** is connected to two first pixels **30**, and the saved space may be used to accommodate other lines or devices, which is beneficial to narrowing the border of the display panel.

[0080] In other embodiments, at least a portion of the first driving circuit **51** is disposed in the second display region **12**, for example, a portion of the first driving circuit **51** may be disposed in the second display region **12**. The first driving circuit **51** may be arranged entirely in the second display region **12**, so that the first display region **11** has an anode and an opening for the pixel to emit light, without the first driving circuit **51**. Providing at least part of the first driving circuit **51** in the second display region **12** is advantageous for saving the space occupied by the first driving circuit **51** in the first display region **11** and for making the border narrower. For example, one first driving circuit **51** may drive at least two first pixels **30** at the same time, one second driving circuit

52 may drive one or more second pixels **40**, at least two first pixels **30** may be driven at the same time via one first driving circuit **51**, at least part of the first driving circuit **51** may be located in the peripheral region or the central region of the second display region **12**, the space occupied by the first driving circuit **51** in the first display region **11** is reduced, and the saved space may be used to accommodate other lines or devices, which is beneficial to increasing the area of the display region and for making the border of the display panel narrower.

[0081] Optionally, the display panel further includes: connection lines **60**, each connection line **60** is connected to the electrodes of at least two first pixels **30**, and each first driving circuit **51** is electrically connected to a corresponding connection line **60**. The connection line **60** may be a metal line, and the material of the metal line may include at least one of aluminum, copper, molybdenum, titanium or silver. For example, the material of the metal line may include molybdenum or silver, allowing the metal line to have a higher conductivity and reducing losses. The connection line **60** may be a separate line, and the connection line **60** may be arranged at a same layer as a gate electrode, or may be located above or below the gate electrode layer, which may be determined according to practical needs. The connection line **60** may be arranged above or below an anode layer; the material of the connection line **60** may be the same with or different from the material of the anode layer. For example, the material of the connection line **60** may be the same as the material of the anode layer, with both the material of the connection line **60** and the material of the anode layer being indium tin oxide (ITO), which has a better light transmittance. Using the same material facilitates processing and manufacturing, simplifying the process. The connection line **60** may serve as the anode layer, and there is no need to separately form the connection line **60**, further simplifying the process. The anode layer serve as a light transmitting material layer, for example, the anode layer may be an indium tin oxide (ITO) layer, which may improve light transmittance and reduce power consumption. For example, each connection line **60** is connected to the electrodes of two first pixels **30**, and each first driving circuit **51** is electrically connected to a corresponding connection line **60**. Each connection line **60** is connected to the electrodes of two first pixels **30** in two adjacent columns respectively, and two first pixels **30** in two adjacent columns respectively connected to the same connection line **60** may be located in the same row, which is beneficial to reducing the length of the connection line **60** and reducing the space occupation.

[0082] Optionally, each connection line **60** is connected to two adjacent first pixels **30**, for example, the two adjacent first pixels **30** connected to each connection line **60** may be located in the same row or the same column, which is beneficial to reducing the length of the connection line **60**, reducing the occupation of space, and saved space may be used to accommodate other lines or devices, which is advantageous in making the border of the display panel narrower.

[0083] In some embodiments, as shown in FIGS. **8** to **11**, each first pixel **30** includes a first sub-pixel **31**, one first driving circuit **51** is configured to first sub-pixels **31** of at least two first pixels **30** simultaneously. For example, each first pixel **30** may include one or more first sub-pixels **31**, the first sub-pixel **31** may be a red pixel, and one first driving circuit **51** may drive the first sub-pixels **31** of two first pixels **30** at the same time. Hence, it is able for the same first driving circuit **51** to drive the first sub-pixels **31** of two first pixels **30** at the same time, which is beneficial to saving the space occupied by the first driving circuit **51** in the first display region **11** and to facilitating a narrower border.

[0084] Optionally, as shown in FIGS. **8** to **11**, each first pixel **30** includes a second sub-pixel **32**, one first driving circuit **51** is configured to simultaneously drive second sub-pixels **32** of at least two first pixels **30**, and the first sub-pixel **31** and the second sub-pixel **32** emit light in different colors. For example, each first pixel **30** may include one or more second sub-pixels **32**, the second sub-pixels **32** may be green pixels, one first driving circuit **51** may drive the second sub-pixels **32** of the two first pixels **30** simultaneously, and the first driving circuit **51** driving the first sub-pixels **31** of the two first pixels **30** simultaneously is not the same as the first driving circuit **51** driving the

second sub-pixels **32** of the two first pixels **30** simultaneously. Hence, it is able for the same first driving circuit **51** to drive the second sub-pixels **32** of the two first pixels **30** at the same time, which is advantageous to saving the space occupied by the first driving circuit **51** in the first display region **11** and facilitating a narrower border.

[0085] Optionally, as shown in FIGS. **8** to **11**, each first pixel **30** includes a third sub-pixel **33**, and one first driving circuit **51** is configured to simultaneously drive third sub-pixels **33** of at least two first pixels **30**, the first sub-pixel **31**, the second sub-pixel **32** and the third sub-pixel **33** emit light in different colors. For example, each first pixel **30** may include one or more third sub-pixels **33**, the third sub-pixels **33** may be blue pixels, one first driving circuit **51** may simultaneously drive the third sub-pixels **33** of the two first pixels **30**, the first driving circuit **51** simultaneously driving the first sub-pixels **31** of the two first pixels **30**, and the first driving circuit **51** simultaneously driving the second sub-pixels **32** of the two first pixels **30** are not the same as the first driving circuit **51** simultaneously driving the third sub-pixels **33** of the two first pixels **30**. Hence, it is able for the same first driving circuit **51** to drive the third sub-pixels **33** of the two first pixels **30** at the same time, which is advantageous to saving the space occupied by the first driving circuit **51** in the first display region **11** and facilitating a narrower border.

[0086] During the application, the first pixels **30** may be in a Delta arrangement, an S-like arrangement, a 2 in 1 arrangement, or any other arrangements such as GGRB or blue Diamond, which may be selected according to actual needs. The connection manner in which one first driving circuit **51** drives two or more first pixels **30** at the same time for different arrangements may be determined according to practical applications.

[0087] In some embodiments, the first pixel **30** includes: a first electrode, a light-emitting unit and a second electrode laminated one on another, the first electrode may be an anode and the second electrode may be a cathode, alternatively, the first electrode may be a cathode, the second electrode may be an anode. One first driving circuit **51** is configured to connect the electrodes in at least two first pixels **30**, one first driving circuit **51** may apply electrical signals for the electrodes in at least two first pixels **30** at the same time, and the light-emitting unit may be driven to emit light via the first electrode and the second electrode. The first driving circuit **51** may include a first active layer, a first source electrode and a first drain electrode connected to the first active layer, and a first gate electrode, where the first source electrode and the first drain electrode may be electrically connected to the electrodes of the first pixel **30**, for example, the first source electrode may be electrically connected to the first electrode, the first drain electrode may be electrically connected to the second electrode, and the first pixel **30** may be controlled to be turned on or off by the first driving circuit **51**.

[0088] The second pixel **40** may include: a third electrode, a light-emitting unit and a fourth electrode laminated one on another, the third electrode may be an anode and the fourth electrode may be a cathode, alternatively, the third electrode may be a cathode, the fourth electrode may be an anode. One second driving circuit **52** may be configured to connect the electrodes in at least one second pixel **40**, and the light-emitting unit may be driven to emit light via the third electrode and the fourth electrode. The second driving circuit **52** may include a second active layer, a second source electrode and a second drain electrode connected to the second active layer, and a second gate electrode, where the second source electrode and the second drain electrode may be electrically connected to the electrodes of the second pixel **40**, for example, the first source electrode may be electrically connected to the third electrode, the first drain electrode may be electrically connected to the fourth electrode, and the second pixel **40** may be controlled to be turned on or off by the second driving circuit **52**. The source and drain electrodes in the first driving circuit **51** and the second driving circuit **52** may be arranged in the same layer, the gate electrodes in the first driving circuit **51** and the second driving circuit **52** may be arranged in the same layer, and the active layers in the first driving circuit **51** and the second driving circuit **52** may be arranged in the same layer, thereby simplifying the process.

[0089] As shown in FIGS. 2 and 3, each connection line **60** is connected to two adjacent first pixels **30**, and each connection line **60** is connected to the two adjacent first pixels **30** through via holes of the pixel regions **62**. For example, as shown in FIG. 3, with regard to a pixel region **62** of a first row located at the top or a middle row located at the middle, a first column is located at the far left side, and one connection line **60** is connected to first sub-pixels **31** of two adjacent first pixels **30** through via holes of the pixel regions **62**, so as to drive the first sub-pixels **31** in the first column and the fourth column at the left side at the same time. Another one connection line **60** is connected to second sub-pixels **32** in the two adjacent first pixels **30** through via holes of the pixel regions **62** so as to drive the second sub-pixels **32** in the second column and the fifth column on the left side at the same time. Still another one connection line **60** is connected to third sub-pixels **33** of two adjacent first pixels **30** through via holes of the pixel regions **62**, so as to drive the third sub-pixels **33** of the third column and the sixth column at the left side simultaneously. For the conventional Delta arrangement, the space for compressing one pixel in the border may narrow the border by about 70 μm -90 μm , and this space may accommodate other lines, branching units or devices, etc. thereby to facilitate the narrow border. As shown in FIG. 3, the first electrode **301** may be an anode, and the first electrode **301** may be in a normal anode arrangement. In a pixel region at the edge, there are only the anode and the opening for the pixel to emit light, with no pixel driving circuit below. This may be applied to a display panel with a circular shape or with rounded corners, so that the panel may have a narrow border.

[0090] As shown in FIG. 7, the leftmost pixel regions **62** are in a first column, one connection line **60** is connected to a sub-pixel of the first pixel **30** in the pixel region **62** of the first column and a sub-pixel of the first pixel **30** in the pixel region **62** of the fourth column through via holes of the pixel regions **62**, so as to simultaneously drive the first sub-pixel of the first pixel **30** in the first column on the left side and the fourth column. Another one connection line **60** is connected to a sub-pixel of the first pixel **30** in the pixel region **62** of the second column and a sub-pixel of the first pixel **30** in the pixel region **62** of the fifth column through via holes of the pixel regions **62**, so as to simultaneously drive a second sub-pixels of the first pixel **30** in the second column and the fifth column on the left side. Still another one connection line **60** is connected to a sub-pixel of the first pixel **30** in the pixel region **62** of the third column and a sub-pixel of the first pixel **30** in the pixel region **62** of the sixth column through via holes of the pixel regions **62**, so as to simultaneously drive third sub-pixels of the first pixel **30** in the third column and the sixth column on the left side. It is able to compress the space of two pixels in the border, in the pixel region at the left edge, there are only the anode and the opening for the pixel to emit light, with no pixel driving circuit below, which may be applied to a rectangular display panel, allowing the panel to have a narrow border. To avoid excessive length of the connection line **60**, there may be one, two or three sub-pixels between two sub-pixels connected by the same connection line **60**.

[0091] As shown in FIGS. 4 and 5, each connection line **60** is connected to two first pixels **30**, and each connection line **60** is connected to two first pixels **30** adjacent to each other through via holes of the pixel regions **62**. For example, as shown in FIG. 4, with regard to pixel regions **62** located in a middle row, the leftmost is a first column, one connection line **60** is connected to first sub-pixels **31** of two adjacent first pixels **30** through via holes of the pixel regions **62**, e.g., the first sub-pixels **31** in a first column and a fourth column on the left side, a track located at the position of the first sub-pixel **31** in the seventh column may be connected to one first driving circuit **51**, the connection line **60** is not electrically connected to the first sub-pixel **31** located in the seventh column, and the first driving circuit **51** is connected to the connection line **60** via the track, thereby driving the first sub-pixels **31** in the first column and the fourth column on the left side at the same time. Another one connection line **60** is connected to the second sub-pixels **32** in two adjacent first pixels **30** through via holes of the pixel regions **62**, e.g., the second sub-pixels **32** in the second column and the fifth column on the left side, so as to drive the second sub-pixels **32** in the second left column and the fifth column at the same time via the same first driving circuit **51**. Still another one

connection line **60** is connected to third sub-pixels **33** of two adjacent first pixels **30** through via holes of the pixel regions **62**, e.g., the third sub-pixels **33** in the third column and the sixth column on the left side, so as to drive the third sub-pixels **33** in the third column and the sixth column on the left side simultaneously via the same first driving circuit **51**. With the above-mentioned arrangement, it is able to compress and obtain the spatial region of two pixels in the border, so that the border may be narrowed by about 140 μm -180 μm , and this space may accommodate more units and lines, thereby to achieve an ultra-narrow border. As shown in FIG. 5, the first electrode **301** may be an anode, and the first electrode **301** may be in a normal anode arrangement, and in two pixel regions at the edge, there are only the anode and the opening for the pixel to emit light, with no pixel driving circuit below. With regard to the narrowing of the left and right borders, when two pixels are driven by one first driving circuit **51** in the left and right borders, the space in an underlying pad region also benefits. This is mainly because corners of the underlying pad shrink towards the display region, allowing the lines to shrink upwards and towards inside of the display region, which is beneficial to the narrowing of the lower border.

[0092] Optionally, the display panel further includes: a detection circuit configured to detect brightness of a pixel, at least part of the detection circuit is located in the first display region **11**, for example, part of the detection circuit is located in the first display region **11**, or the detection circuit is entirely located in the first display region **11**. At least part of the detection circuit may be located in the space saved in the first display region **11** due to the first driving circuit **51**, at least part of the detection circuit may overlap the first pixel **30**, and at least part of the detection circuit may be located below the first pixel **30**. The detection circuit may be placed in the first display region **11**, so that the detection circuit is closer to the display region, which is beneficial to save the space occupied by the detection circuit in a corresponding region of the frame body, thereby to provide a narrower border.

[0093] Optionally, as shown in FIG. 6, the display panel further includes: a branching unit **61**, the first driving circuit **51** and the first pixel **30** are electrically connected via the branching unit **61**, so as to facilitate the electrical connection between the first driving circuit **51** and the first pixel **30**, and different branching lines in the branching unit **61** may connect different first pixels **30**. The first driving circuit **51** is electrically connected to the first pixels **30** via the branching unit **61**, the first pixels **30** are electrically connected to the branching unit **61** via the connection lines **60**, each connection line **60** may be connected to two adjacent first pixels **30**, and each branching line in the branching unit **61** may be connected to two adjacent first pixels **30**, thereby facilitating the arrangement of lines while facilitating the reduction of space occupation.

[0094] Optionally, at least part of the branching unit **61** is located in the first display region **11**, for example, a part of the branching unit **61** may be located in the first display region **11**, the space saved by the first driving circuit **51** in the first display region **11** may be used to accommodate the branching unit **61**, and the space occupied by the branching unit **61** in a corresponding region of the frame body may be reduced, thereby to provide a narrower border.

[0095] Optionally, the first display region **11** includes a first region **111**, a second region **112**, a third region **113** and a fourth region **114**, the first region **111**, the second region **112**, the third region **113** and the fourth region **114** are distributed along the periphery of the second display region **12**, and the first pixels are arranged in at least one of the first region **111**, the second region **112**, the third region **113** and the fourth region **114**. When closing to the region where the first pixels **30** are distributed, it is advantageous to making the border narrower.

[0096] Optionally, a distribution density of the first pixels **30** in the first display region **11** is less than a distribution density of the second pixels **40** in the second display region **12**. Since the distribution density of the first pixels **30** in the first display region **11** is less than the distribution density of the second pixels **40** in the second display region **12**, the quantity of the first driving circuits required for driving the first pixels **30** in the first display region **11** is reduced, the space occupied by the first driving circuits **51** is reduced, and the saved space may be used to

accommodate other lines or devices, which is beneficial to increasing the area of the display region and narrowing the border of the display panel.

[0097] The display device according to the embodiments of the present disclosure, including the display panel described in the above-described embodiments, which may make the frame body narrower and facilitate miniaturization.

[0098] The display device further includes: a frame body covering the non-display region **20**. The frame body may be arranged in a border region **21** of the non-display region **20**, and the border region **21** may surround the periphery of the first display region **11**. Since the space saved by the first driving circuit **51** in the first display region **11** may be used to accommodate other lines or devices, it is able to reduce the space occupation for the non-display region **20** and make the frame body narrower.

[0099] The embodiments of the present disclosure are described above with reference to the accompanying drawings. However, the present disclosure is not limited to the foregoing specific implementations. The foregoing specific implementations are merely illustrative rather than limitative. In light of the teachings of the present disclosure, a person of ordinary skill in the art may further make various forms without departing from the spirit of the present disclosure and the scope of the claims, and these forms all fall within the scope of the present disclosure.

Claims

1. A display panel, comprising a display region and a non-display region, wherein the non-display region surrounds the display region, the display region comprises a first display region and a second display region, the first display region surrounds the periphery of the second display region, a plurality of first pixels is arranged in the first display region, a plurality of second pixels is arranged in the second display region, the display region comprises a first driving circuit configured to drive the first pixels and a second driving circuit configured to drive the second pixels; and one first driving circuit is configured to simultaneously drive at least two first pixels.
2. The display panel according to claim 1, wherein one first driving circuit is configured to simultaneously drive two first pixels in two adjacent columns respectively.
3. The display panel according to claim 2, wherein two first pixels respectively in two columns driven simultaneously by a same first driving circuit are in a same row.
4. The display panel according to claim 1, wherein at least part of the first driving circuit is arranged in the second display region.
5. The display panel according to claim 1, further comprising: connection lines, wherein each of the connection lines is connected to electrodes of at least two first pixels, and each first driving circuit is electrically connected to a corresponding one of the connection lines.
6. The display panel according to claim 5, wherein each of the connection lines is connected to two adjacent first pixels.
7. The display panel according to claim 1, wherein each of the first pixels comprises a first sub-pixel, and one first driving circuit is configured to simultaneously drive first sub-pixels of at least two first pixels.
8. The display panel according to claim 7, wherein each of the first pixels comprises a second sub-pixel, and one first driving circuit is configured to simultaneously drive second sub-pixels of at least two first pixels, wherein the first sub-pixel and the second sub-pixel emit light in different colors.
9. The display panel according to claim 1, wherein the first pixel comprises: a first electrode, a light-emitting unit and a second electrode laminated one on another, and one first driving circuit is configured to connect electrodes in at least two first pixels.
10. The display panel according to claim 1, further comprising: a detection circuit configured to detect brightness of a pixel, at least part of the detection circuit being located in the first display

region.

- 11.** The display panel according to claim 1, further comprising: a branching unit, wherein the first driving circuit is electrically connected to the first pixels via the branching unit.
 - 12.** The display panel according to claim 11, wherein at least part of the branching unit is located in the first display region.
 - 13.** The display panel according to claim 1, wherein the first display region comprises a first region, a second region, a third region and a fourth region, the first region, the second region, the third region and the fourth region are distributed along the periphery of the second display region, and the first pixels are arranged in at least one of the first region, the second region, the third region and the fourth region.
 - 14.** The display panel according to claim 1, wherein a distribution density of the first pixels in the first display region is less than a distribution density of the second pixels in the second display region.
 - 15.** A display device comprising the display panel according to claim 1; wherein the display device further comprises: a frame body covering the non-display region.
 - 16.** The display device according to claim 15, wherein one first driving circuit is configured to simultaneously drive two first pixels in two adjacent columns respectively.
 - 17.** The display device according to claim 15, wherein at least part of the first driving circuit is arranged in the second display region.
 - 18.** The display device according to claim 15, wherein the display panel further comprises: connection lines, wherein each of the connection lines is connected to electrodes of at least two first pixels, and each first driving circuit is electrically connected to a corresponding one of the connection lines.
 - 19.** The display device according to claim 15, wherein the display panel further comprises: a branching unit, wherein the first driving circuit is electrically connected to the first pixels via the branching unit.
 - 20.** The display device according to claim 15, wherein a distribution density of the first pixels in the first display region is less than a distribution density of the second pixels in the second display region.
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